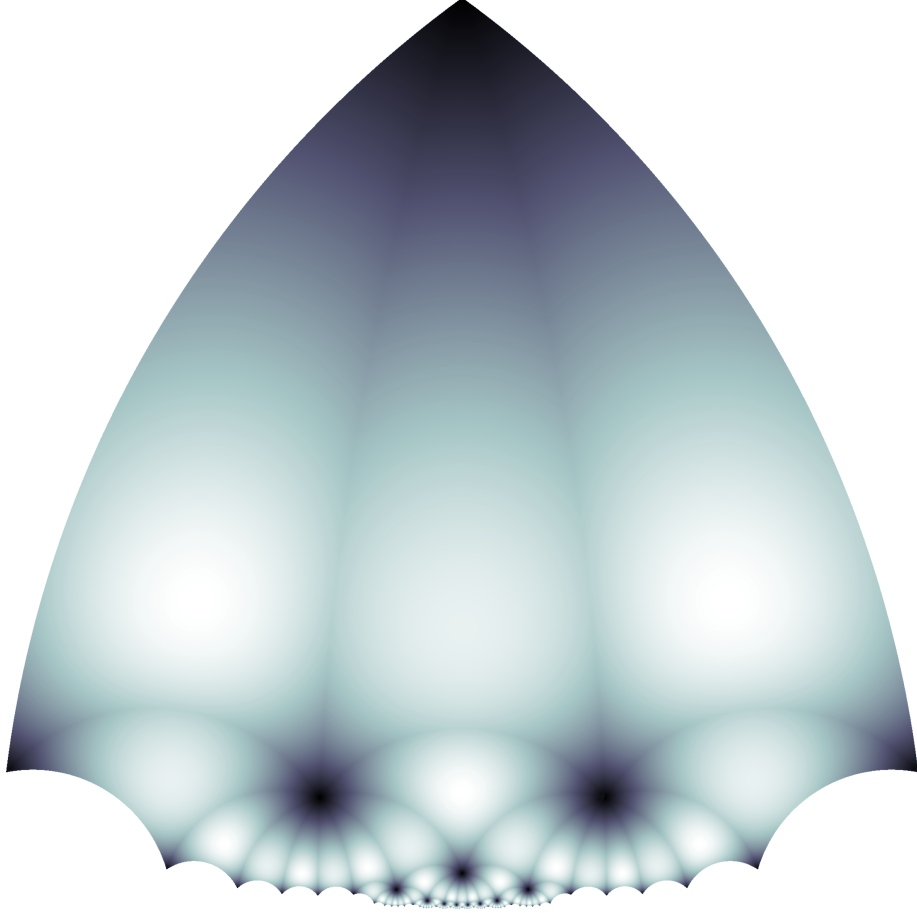


Binomial stabilizer states

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Abstract

The classically simulable Clifford group action, on the space of odd prime stabilizer states Φ , extends to a deformation of the space by a complex parameter τ . Deformed states $\Phi(\tau)$ are called binomial stabilizer states. They are modular functions: the deformation parameter τ lives on the universal cover of a Fermat curve. If stabilizer states are discrete Gaussians, then binomial stabilizer states are discrete harmonic oscillator wavefunctions.

Odd prime qudits

An odd prime qudit is an odd prime dimensional Hilbert space $\mathbf{C}^p$. Here $\mathbf{C}$ is the field of complex numbers and p is an odd prime number. A qudit state Φ is a vector in $\mathbf{C}^p$. We will write Φ like

$$\Phi = \sum_{n \in \mathbf{Z}/p} A_n X^n$$

where the symbol X^n is the standard n th basis vector in $\mathbf{C}^p$ and the amplitude A_n is a number in $\mathbf{C}$.

The Pauli group and stabilizer states

The automorphism group of the qudit $\mathbf{C}^p$ is the special unitary group $SU(p)$. The Pauli group is a finite subgroup of $SU(p)$. It is generated by the linear operators Z, X acting on the basis vectors like

$$ZX^n = q^n X^n \quad XX^n = X^{n+1}$$

where q is a primitive p th root of unity. The generators have order p and they satisfy the quantum commutation relation

$$ZX = qXZ.$$

A stabilizer state is a qudit state that is fixed by an element of the Pauli group.

The quantum torus

The quantum torus $\mathbf{T}$ is the polynomial algebra generated by Z, X . The Pauli group is the monomial subgroup of $\mathbf{T}$.

Binomial operators and Fermat curves

An operator

$$W = U(S + TV)$$

in the quantum torus $\mathbf{T}$ is a cyclic binomial operator if $U, V \in \mathbf{T}$ have order p and satisfy the quantum commutation relation $UV = qVU$, and $S, T \in \mathbf{C}$ is a point on the affine Fermat curve

$$S^p + T^p = 1. \tag{1}$$

A cyclic binomial operator acts on qudits like the Hamiltonian of a harmonic oscillator.

Theorem 1. A cyclic binomial operator has order p .

Proof. The commuting operators $V_k = (1 + q^k TV/S)$ interact with U like $V_k U = U V_{k-1}$ so

$$W^p = (SUV_0)^p = S^p \prod_{k \in \mathbf{Z}/p} V_k.$$

Take the logarithm of the product

$$\begin{aligned} \log \prod_{k \in \mathbf{Z}/p} V_k &= \sum_{n \in \mathbf{Z}_+} \sum_{k \in \mathbf{Z}/p} (-1)^n \frac{(q^k TV/S)^n}{n} \\ &= \sum_{n \in \mathbf{Z}_+} (-1)^n \frac{(TV/S)^n}{n} \sum_{k \in \mathbf{Z}/p} q^{kn} \\ &= \sum_{n \in \mathbf{Z}_+} (-1)^n \frac{(TV/S)^{pn}}{n} \\ &= \sum_{n \in \mathbf{Z}_+} (-1)^n \frac{(T^p/S^p)^n}{n} \\ &= \log(1 + T^p/S^p) \end{aligned}$$

then exponentiate $W^p = S^p + T^p = 1$. ■

Gauss binomial coefficients

The quantum Pochhammer symbol

$$(x; q)_n = \prod_{k=0}^{n-1} (1 - xq^k)$$

is a degree n polynomial in a formal variable x with coefficients in the integer polynomial ring $\mathbf{Z}[q]$. Write $(x)_n = (x; q)_n$ when q is fixed. The sequence $(x)_1, (x)_2, (x)_3 \dots$ is uniquely determined by the recursion

$$(x)_{n+1} = (1 - xq^n) (x)_n \tag{2}$$

with the initial condition $(x)_1 = 1 - x$. The quantum binomial theorem

$$(x)_n = \sum_{k=0}^n (-1)^k q^{k(k-1)/2} \binom{n}{k}_q x^k \tag{3}$$

says that the coefficients of the quantum Pochhammer symbols are essentially Gauss binomial coefficients

$$\binom{n}{k}_q = \frac{(q)_n}{(q)_k (q)_{n-k}}.$$

Cyclic Pochhammer symbols

The cyclic Pochhammer symbol Ω_n is defined by the formula

$$\Omega_n(S, T; q) = (-T/S; q)_n S^n$$

where $S, T \in \mathbf{T}$ are commuting operators on the degree p Fermat curve 1 and q is a primitive p th root of unity.

Cyclic Pochhammer symbols appear in formulas for the Boltzmann weights of a solvable three-dimensional lattice model (1993), and also in the definition of the quantum dilogarithm at roots of unity (1994). The index n is well defined mod p and $\Omega_0 = 1$.

The values at n and $-n$ are related by the formula

$$\Omega_n(S, T; q) \cdot \Omega_{-n}(T, qS; q) = q^{n(n-1)/2}.$$

The values at q and q^{-1} are related by the formula

$$\Omega_n(S, T; q) / \Omega_n(T, S; q^{-1}) = q^{n(n-1)/2}.$$

The Fourier transform of the sequence $\Omega_0, \Omega_1, \Omega_2 \dots$ is the sequence $F_0, F_1, F_2 \dots$ where

$$F_m = \sum_{n \in \mathbf{Z}/p} \Omega_n q^{mn}.$$

Use the Pochhammer recursion 2 to write

$$\begin{aligned} F_m &= q^m \cdot \sum_{n \in \mathbf{Z}/p} (S + T q^n) \Omega_n q^{mn} \\ &= q^m \cdot (S F_m + T F_{m+1}) \end{aligned}$$

then rearrange terms

$$F_{m+1} = ((-S/T) + (1/T)q^{-m}) \cdot F_m$$

to see that $F_m(q)$ satisfies the defining functional equation for $\Omega_m(q^{-1})$.

The Fourier transform of the cyclic Pochhammer symbols is

$$F(\Omega(S, T; q)) = K \cdot \Omega(-S/T, 1/T; q^{-1})$$

where $K = \sum_n \Omega_n(S, T; q)$.

Binomial stabilizer states

A binomial stabilizer state is a qudit state that is fixed by a cyclic binomial operator. It is like a qudit harmonic oscillator wavefunction.

Binomial stabilizer states deform Pauli stabilizer states with a complex parameter in the universal cover of the degree p Fermat curve 1. Stabilizer states live at the vertices of the equilateral triangle lattice that covers the ramification locus of the Belyi map $(S, T) \mapsto S^p$.

Theorem 2. Choose some quantum commuting monomial operators $U = Z^a X^b$ and $V = Z^c X^d$. The binomial operator $W = U(S + TV)$ fixes the qudit state with the amplitudes

$$A_n = \sum_{m \in \mathbf{Z}/p} H_{m,n}(q^{ab/2}S, q^{(ab+cd+1)/2}T) q^{-abm^2/2+amn}$$

where

$$H_{m,n}(S, T) = \sum_{k=0}^m \binom{m}{k}_q T^k S^{m-k} q^{(1-cd)k^2/2+(cn-adm)k}$$

is like a Hermite polynomial.

Proof. The Fourier transform

$$F_n W = \frac{1}{p} \sum_{m \in \mathbf{Z}/p} q^{mn} W^m$$

maps qudit states to eigenvectors of W with eigenvalue q^n . This follows from the order of W that we determined in Theorem 1. So the binomial stabilizer state is $\Phi = F_0 W F_0 X$. Let $V_k = (S + q^k TV)$ so that

$$\begin{aligned} W^m &= (UV_0)^m \\ &= \left(\prod_{k=1}^m V_k \right) U^m \\ &= \Omega_m(S, qTV) U^m \\ &= \sum_{k=0}^m \binom{m}{k}_q T^k S^{m-k} q^{k(k+1)/2} V^k U^m \end{aligned}$$

by the quantum binomial theorem 3. Expand $W^m X^n$ in X, q and sum over m, n to isolate the amplitudes of Φ . $\blacksquare$

When $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the identity matrix, the binomial stabilizer states are

$$\begin{aligned} \Phi &= \frac{1}{p} \sum_{m,n \in \mathbf{Z}/p} q^{mn} \Omega_m(S, T; q^{-1}) X^n \\ &= K \sum_{n \in \mathbf{Z}/p} \Omega_{-n}(-S/T, 1/T) X^n \\ &= K \sum_{n \in \mathbf{Z}/p} \frac{q^{n(n+1)/2}}{(qS)_n} T^n X^n \end{aligned}$$

where $K = \frac{1}{p} \sum_n \Omega_n(S, T; q^{-1}) S^n$. At the ramification points $S = 1$, the cyclic Pochhammer symbols are trivially $\Omega_n = 1$, and the binomial stabilizer state is the Pauli stabilizer state $\Phi = X^0$.

The Wigner function and mana

The Wigner function G of a qudit state $\Phi = \sum_n A_n X^n$ is the real $p \times p$ matrix

$$G_{ij}(\Phi) = \frac{1}{q^{2ijp}} \sum_{n \in \mathbf{Z}/p} q^{2jn} A_n A_{2i-n}^*. \quad (2006)$$

Here A^* is the complex conjugate of the amplitude A .

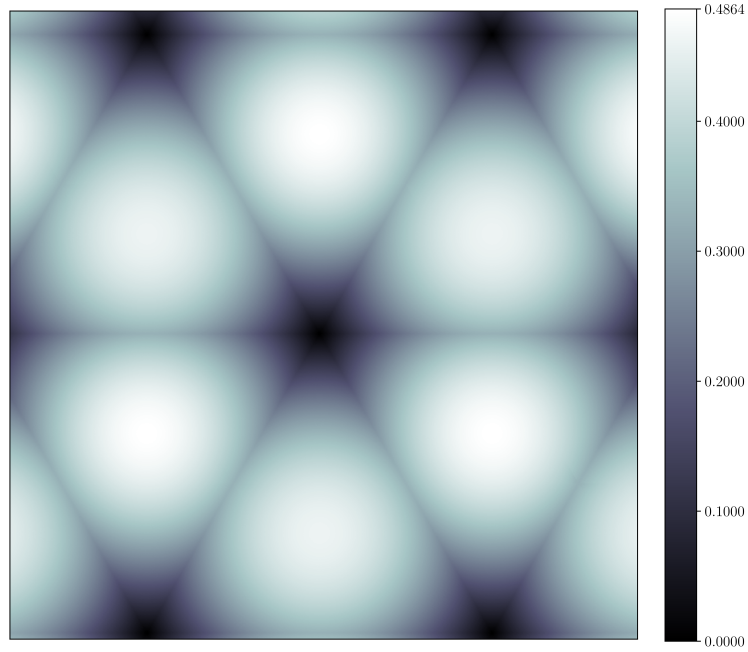
For a fixed odd prime number p , the Wigner function restricted to the class of binomial stabilizer states pulls back to a matrix valued function on the universal cover of the degree p Fermat curve 1. The universal cover is Euclidean when $p = 3$. Otherwise it is hyperbolic.

The mana of Φ is the nonnegative real number

$$M(\Phi) = \log \sum_{i,j \in \mathbf{Z}/p} |G_{ij}(\Phi)|. \quad (2014)$$

It is a measure of complexity. It vanishes on Pauli stabilizer states. It is invariant under Clifford gates. The maximum of the mana for the binomial stabilizer states for $p = 3$ and $p = 5$ is about 0.48 and 0.64 .

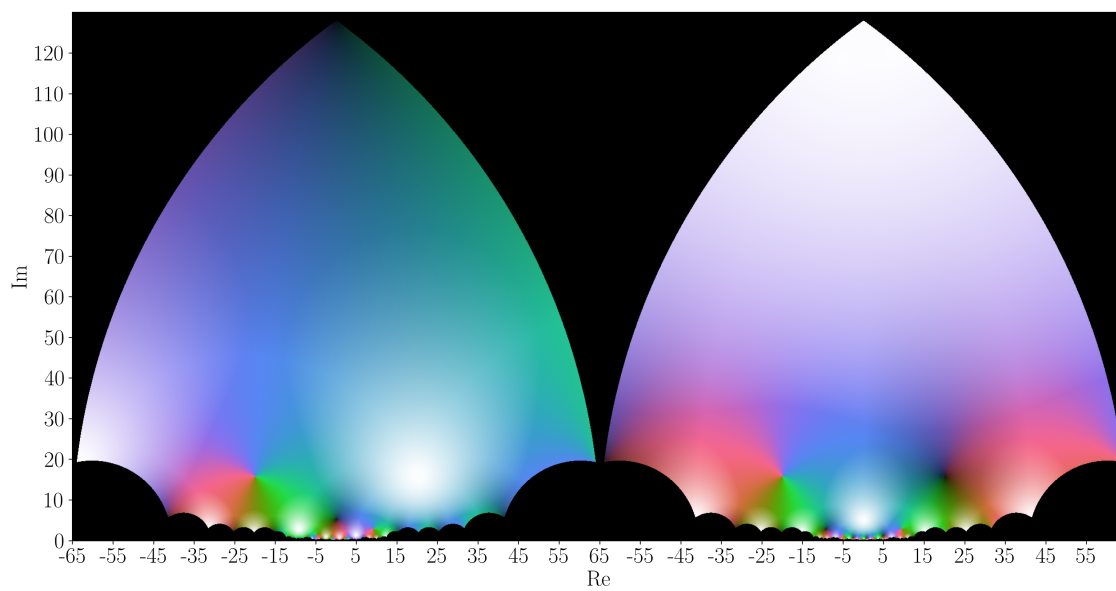
Plots



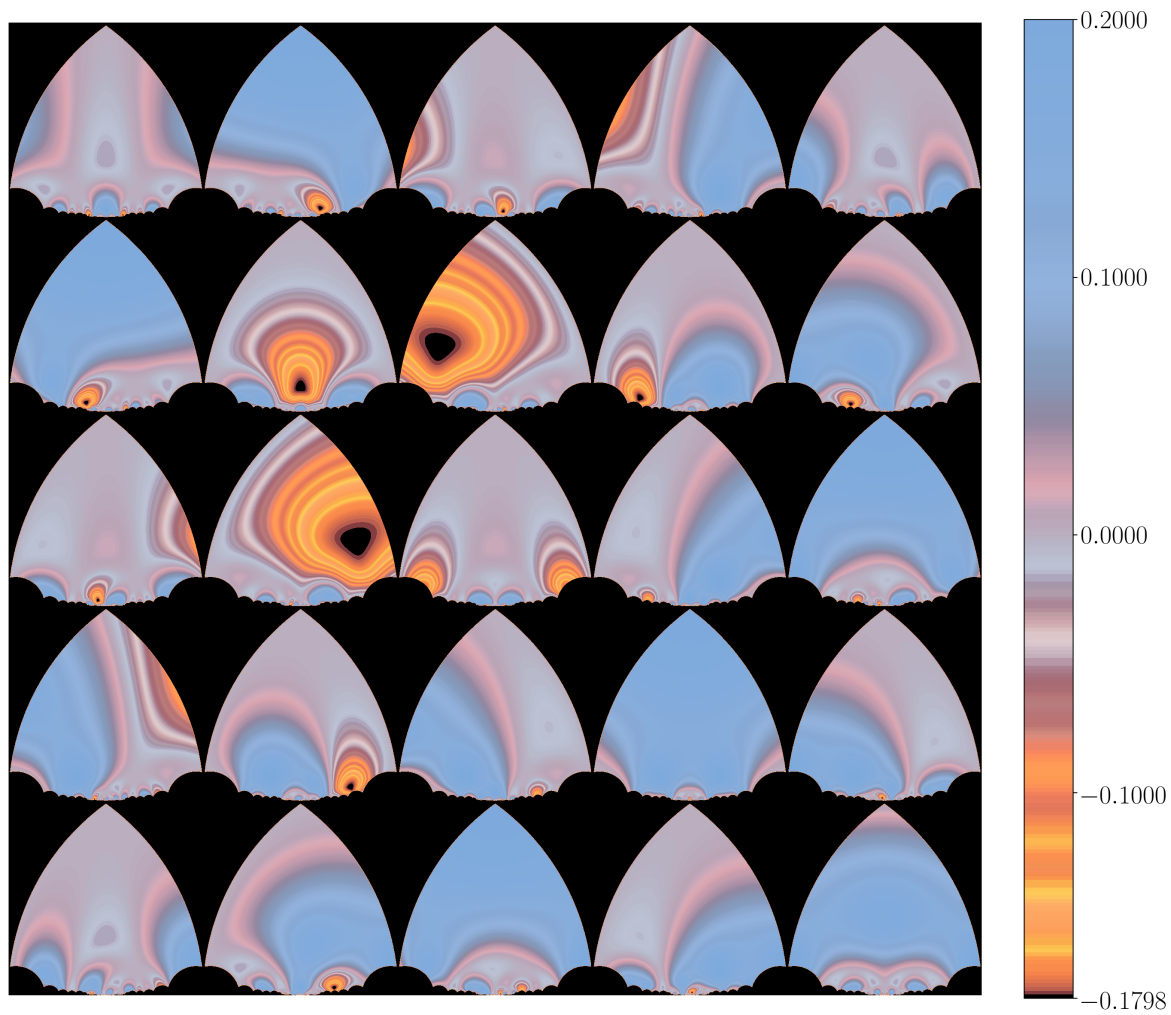
The mana of the qutrit binomial stabilizer states, as a function of the complex plane.



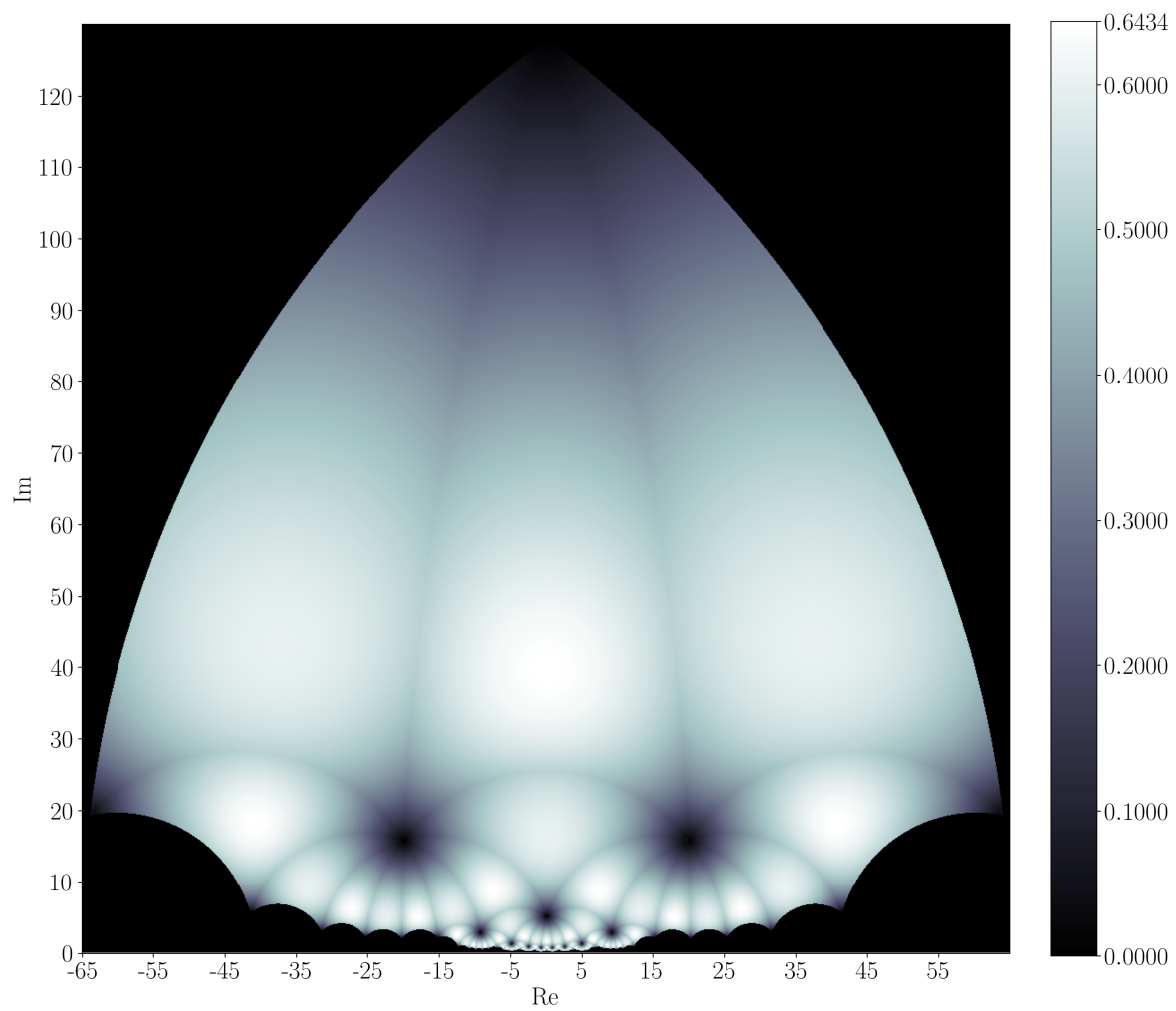
The complex plane.



S, T coordinates on the degree 5 Fermat curve.



The Wigner function evaluated on the ququint binomial ground states.



The mana of the quint binomial stabilizer states, as a function of the upper half plane.

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